

CS 350 - Software Development

Professor: Prof. Alex Rivera | arivera@university.edu | Engineering 312

Office Hours: Mon 5:00-6:30 PM, Wed 1:00-2:30 PM, or by appointment

Term: Fall 2026 | Credits: 3

Location: See schedule below

Course Description

A project-oriented course on professional software engineering practices. Students work in teams of 3-4 using Agile/Scrum methodology to build a full-stack web application over three sprints. Topics include: version control with Git/GitHub, test-driven development (TDD), CI/CD pipelines (GitHub Actions), containerization (Docker), REST API design, design patterns, and code review best practices. Lecture: Mon/Wed 3:30-4:45 PM in Engineering 102. Lab: Friday 2:00-3:50 PM in Engineering Computer Lab 110.

Learning Objectives

Students will:

1. Apply Agile/Scrum practices including sprint planning, standups, and retrospectives.
2. Use Git branching strategies and code review workflows professionally.
3. Write unit, integration, and end-to-end tests using pytest and Playwright.
4. Build and deploy a containerized web application with Docker and GitHub Actions.
5. Design and implement a clean REST API following OpenAPI specification.
6. Conduct and receive structured code reviews using industry-standard rubrics.

Required Materials

No required textbook. Weekly readings posted on course portal (free/open access).

Software: Git, Python 3.11+, Node.js 20+, Docker Desktop, VS Code (recommended).

Required accounts: GitHub (student pro account - free), GitHub Actions minutes provided.

Recommended: 'Clean Code' by Robert C. Martin (optional but highly useful).

Assignments & Due Dates

Lab 1 - Git Workflow and Code Review - Due: September 14, 2026 (5%)

Fork the starter repository, implement a small feature using a feature branch, open a pull request with a detailed description, and review a classmate's PR using the provided rubric. Submit GitHub PR link on course portal by 11:59 PM.

Lab 2 - Test-Driven Development - Due: September 28, 2026 (5%)

Implement 3 small functions using the red-green-refactor TDD cycle with pytest. Achieve 90%+ code coverage. Submit pytest report and code on GitHub by 11:59 PM.

Sprint 1 Demo - MVP Feature Set - Due: October 5, 2026 (10%)

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Live 10-minute demo of your team's Sprint 1 deliverable in lab. Must include: at least 3 working user stories, a passing CI pipeline, and a deployed demo environment. Submit Sprint 1 retrospective document on course portal same day.

Lab 3 - Docker and CI/CD - Due: October 12, 2026 (5%)

Containerize a provided Flask application using Docker. Write a GitHub Actions workflow that builds the image, runs tests, and publishes to GitHub Container Registry. Submit docker-compose.yml, Dockerfile, and workflow YAML on GitHub.

Midterm Exam - Due: October 19, 2026 (15%)

Written exam on software engineering concepts: Git internals, testing theory, design patterns (SOLID principles, Factory, Observer, Strategy), REST API design, and Agile principles. Open-notes (personal notes only, not slides). 75 minutes.

Sprint 2 Demo - Core Feature Complete - Due: November 2, 2026 (15%)

15-minute live demo. Sprint 2 must add: database persistence, user authentication, REST API endpoints, and full test coverage (>80%). Code review from another team due at same time. Submit Sprint 2 retrospective and updated backlog.

Lab 4 - REST API Design - Due: November 9, 2026 (5%)

Design and implement a REST API for a given domain model. Write an OpenAPI 3.0 specification, implement endpoints in FastAPI, and write integration tests. Submit on GitHub by 11:59 PM.

Code Review Assignment - Due: November 16, 2026 (5%)

Conduct a structured code review of a provided codebase (400-line Python module). Identify: bugs, security issues, performance bottlenecks, style violations, and test gaps. Submit written review (minimum 600 words) on course portal.

Sprint 3 Demo - Final Polish - Due: November 30, 2026 (15%)

20-minute final sprint demo. Must include: complete feature set, end-to-end tests with Playwright, full CI/CD pipeline, Docker Compose for local dev, and production deployment (Render or Railway free tier). All team members must present. Submit Sprint 3 retrospective.

Final Project Submission - Due: December 7, 2026 (10%)

Final code submission including: clean README, architecture diagram, API docs, test results, and individual contribution summary. Tag release v1.0.0 on GitHub and submit the tag link on the course portal by 11:59 PM.

Final Exam - Due: December 17, 2026 (10%)

Comprehensive written final covering all course topics. 90 minutes. Emphasis on design patterns, system design, and trade-off analysis. Open-notes (one 8.5x11 sheet, both sides).

Course Schedule Notes

Week 1 (Sep 8-12): Course intro, Git review, team formation

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Week 2 (Sep 15-19): Agile/Scrum, user stories, backlog grooming
Week 3 (Sep 22-26): TDD, pytest, mocking [Sprint 1 begins]
Week 4 (Sep 29-Oct 3): CI/CD, GitHub Actions basics
Week 5 (Oct 6-10): Docker, containers, docker-compose
Week 6 (Oct 13-17): Design patterns I + MIDTERM EXAM (Oct 19)
Week 7 (Oct 20-24): REST API design, OpenAPI; SPRINT 1 DEMO (Oct 5)
Week 8 (Oct 27-31): Databases, ORMs, migrations [Sprint 2 begins]
Week 9 (Nov 3-7): Authentication, security basics
Week 10 (Nov 10-14): Design patterns II, code review techniques
Week 11 (Nov 17-21): End-to-end testing, Playwright [Sprint 3 begins]
Week 12 (Nov 24): Deployment, monitoring [Monday only]
Week 13 (Dec 1-5): Performance, refactoring, tech debt
Week 14 (Dec 7-10): Presentations and project demos
FINAL EXAM: December 17, 2026

Grading Policy

Labs (4): 20%
Sprint Demos (3): 40%
Code Review: 5%
Final Project: 10%
Midterm Exam: 15%
Final Exam: 10%

Team grades are adjusted by peer evaluation scores (+/-10%). Late labs lose 20% per day. Sprint demos cannot be submitted late. Grade scale: A 93+, A- 90, B+ 87, B 83, B- 80, C+ 77, C 73, D 60, F <60.